PROOFS OF INTEGRALITY OF BINOMIAL COEFFICIENTS

KEITH CONRAD

1. Introduction

The *binomial coefficients* are the numbers

$$\binom{n}{k} := \frac{n!}{k!(n-k)!} = \frac{n(n-1)\cdots(n-k+1)}{k!} \tag{1.1}$$

where $0 \leq k \leq n$ in $\mathbf{Z}$. Their name comes from their appearance in the binomial theorem

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}. \tag{1.2}$$

but we will use (1.1), not (1.2), as their definition. While $\binom{n}{k}$ is, by (1.1), obviously a positive rational number, it is not obvious from (1.1) that $\binom{n}{k}$ is an integer.

Theorem 1.1. *For all integers n and k with $0 \leq k \leq n$, $\binom{n}{k} \in \mathbf{Z}$.*

We will give six proofs of Theorem 1.1 and then generalize binomial coefficients to q-binomial coefficients, which have an analogue of Theorem 1.1.[1]

2. Proof by Combinatorics

Our first proof will be a proof of the binomial theorem that, at the same time, gives binomial coefficients a combinatorial meaning. By the distributive law,

$$(1+x)^n = \underbrace{(1+x)(1+x)\cdots(1+x)}_{n \text{ terms}} = \sum_{k=0}^{n} c_{n,k} x^k,$$

where $c_{n,k}$ is the number of ways to pick k items out of n items: in k of the factors $1+x$ of $(1+x)^n$ pick x, and in the other factors pick 1. By this description, $c_{n,k} \in \mathbf{Z}^+$ and $c_{n,0} = 1$.

To get a formula for $c_{n,k}$, we have

$$\begin{aligned}
(1+x)^{n+1} &= (1+x)(1+x)^n \\
&= (1+x)\sum_{k=0}^{n} c_{n,k} x^k \\
&= \sum_{k=0}^{n} c_{n,k} x^k + \sum_{k=0}^{n} c_{n,k} x^{k+1} \\
&= 1 + \sum_{k=1}^{n} (c_{n,k} + c_{n,k-1}) x^k + x^{n+1},
\end{aligned}$$

[1]Another generalization is $\frac{n!}{k_1!k_2!\cdots k_d!} \in \mathbf{Z}$ when $k_1 + k_2 + \cdots + k_d = n$ with $d \geq 1$ and all k_i in $\mathbf{N}$.

and differentiation of both sides gives us

$$(n+1)(1+x)^n = \sum_{k=1}^{n} k(c_{n,k} + c_{n,k-1})x^{k-1} + (n+1)x^n.$$

Equating coefficients of x^{k-1} on both sides for $k = 1, \ldots, n-1$, we get

$$(n+1)c_{n,k-1} = k(c_{n,k} + c_{n,k-1}) \Longrightarrow c_{n,k} = \frac{n-(k-1)}{k} c_{n,k-1}. \tag{2.1}$$

Iterating this a second time,

$$c_{n,k} = \frac{(n-(k-1))(n-(k-2))}{k(k-1)} c_{n,k-2}.$$

Continuing, we eventually get

$$c_{n,k} = \frac{(n-k+1)(n-k+2)\cdots(n-k+k)}{k(k-1)\cdots(1)} c_{n,k-k} = \frac{(n-k+1)(n-k+2)\cdots n}{k!} = \binom{n}{k},$$

so $\binom{n}{k}$ is a positive integer. (There are other ways to show $c_{n,k} = n!/k!(n-k)!$ by counting.)

3. Proof by Recursion

Binomial coefficients are determined by the Pascal's triangle recursion, illustrated below.

$$\begin{array}{ccccccccccccccccccc}
 & & & & & & 1 & & & & & & & & \binom{0}{0} & & & & \\
 & & & & 1 & & & & 1 & & & & & \binom{1}{0} & & \binom{1}{1} & & & \\
 & & & 1 & & 2 & & & 1 & & & \binom{2}{0} & & \binom{2}{1} & & \binom{2}{2} & & \\
 & & 1 & & 3 & & 3 & & 1 & & \binom{3}{0} & & \binom{3}{1} & & \binom{3}{2} & & \binom{3}{3} & \\
 & 1 & & 4 & & 6 & & 4 & & 1 & \binom{4}{0} & & \binom{4}{1} & & \binom{4}{2} & & \binom{4}{3} & & \binom{4}{4}
\end{array}$$

Algebraically, the recursion is $\binom{n}{0} = \binom{n}{n} = 1$ for $n \geq 0$, and

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}. \tag{3.1}$$

for $1 \leq k \leq n$. The reader can check directly that (1.1) satisfies this recursion.

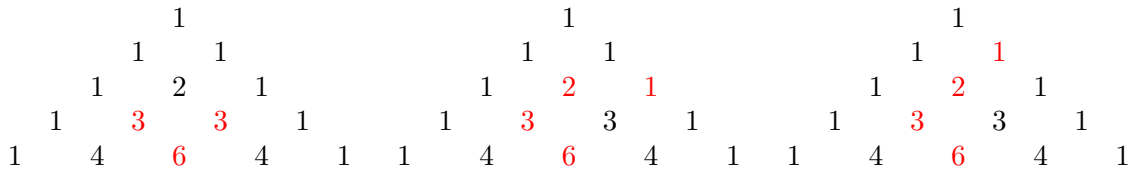
For $k \leq n-1$ we can apply the recursion to the second term on the right in (3.1), getting

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-2}{k-1} + \binom{n-2}{k}.$$

Continuing,

$$\begin{aligned}
\binom{n}{k} &= \binom{n-1}{k-1} + \binom{n-2}{k-1} + \binom{n-3}{k-1} + \cdots + \binom{k-1}{k-1} \\
&= \sum_{j=0}^{n-k} \binom{n-1-j}{k-1} \\
&= \sum_{j=k-1}^{n-1} \binom{j}{k-1}. \qquad (3.2)
\end{aligned}$$

This says each entry in Pascal's triangle is not just the sum of the two entries directly above it, but is the sum of the entries in all previous rows shifted over to the left by 1 position. The diagram below illustrates how $\binom{4}{2}$ (where $k = 2$) is the sum of $\binom{a}{1}$ for $a = 1, 2, 3$.



Since $\binom{n}{k}$ is a sum of binomial coefficients with denominator $k-1$, if all binomial coefficients with denominator $k-1$ are in $\mathbf{Z}$ then so are all binomial coefficients with denominator k, by (3.2). Thus the integrality of all $\binom{n}{k}$ is proved by induction since it is clear when $k = 0$.

4. Proof by Calculus

For $|x| < 1$ we have the geometric series expansion

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots = \sum_{k \geq 0} x^k.$$

There is no obvious connection between this and binomial coefficients, but we will discover one by looking at the series expansion of powers of $1/(1-x)$. For $m \geq 1$,

$$\frac{1}{(1-x)^m} = \left(\frac{1}{1-x}\right)^m = (1 + x + x^2 + x^3 + \cdots)^m = \sum_{k \geq 0} a_{m,k} x^k,$$

where $a_{m,k} \in \mathbf{Z}^+$ by the way power series (like polynomials) multiply. Working out the first few powers of $1/(1-x)$, we get the expansions shown below.

$$\begin{aligned}
1/(1-x) &= 1 + x + x^2 + x^3 + x^4 + x^5 + \cdots \\
1/(1-x)^2 &= 1 + 2x + 3x^2 + 4x^3 + 5x^4 + \cdots \\
1/(1-x)^3 &= 1 + 3x + 6x^2 + 10x^3 + 15x^4 + \cdots \\
1/(1-x)^4 &= 1 + 4x + 10x^2 + 20x^3 + 35x^4 + \cdots \\
1/(1-x)^5 &= 1 + 5x + 15x^2 + 35x^3 + 70x^4 + \cdots
\end{aligned}$$

There are binomial coefficients along diagonals. For instance, the coefficients of degree 0, 1, 2, 3, 4 in $1/(1-x)^5, 1/(1-x)^4, \ldots, 1/(1-x)$ are (tilt your head left) 1, 4, 6, 4, 1. This suggests trying to describe $a_{m,k}$ as an explicit binomial coefficient. Then, since we already know each $a_{m,k}$ is an integer, we will get a proof that binomial coefficients are integers.

Taylor's formula says $a_{m,k} = f^{(k)}(0)/k!$ where $f(x) = 1/(1-x)^m = (1-x)^{-m}$. Successive differentiation of $f(x) = (1-x)^{-m}$ with the power rule and chain rule gives

$$\begin{aligned}
f'(x) = (-m)(1-x)^{-m-1}(-1) &= m(1-x)^{-m-1}, \\
f''(x) = (-m-1)m(1-x)^{-m-2}(-1) &= (m+1)m(1-x)^{-m-2}, \\
f'''(x) = (-m-2)(m+1)m(1-x)^{-m-3}(-1) &= (m+2)(m+1)m(1-x)^{-m-3},
\end{aligned}$$

and in general for $k \geq 0$

$$f^{(k)}(x) = (m+k-1)\cdots(m+2)(m+1)m(1-x)^{-m-k},$$

so $f^{(k)}(0) = (m+k-1)\cdots(m+2)(m+1)m$. Thus

$$a_{m,k} = \frac{(m+k-1)\cdots(m+1)m}{k!} \stackrel{!}{=} \binom{m+k-1}{k}.$$

Now if we *start* with integers $n \geq k \geq 0$, let's make $\binom{n}{k}$ be $\binom{m+k-1}{k}$ by using $m = n-k+1$. This is a positive integer since $n \geq k$. Therefore $\binom{n}{k} = a_{n-k+1,k} \in \mathbf{Z}^+$.

5. Proof by Number Theory

We observed in the introduction that $\binom{n}{k}$ is a positive rational number. We will prove it is an integer by showing its denominator is 1 using prime factorizations.

The *multiplicity* of a prime p in a nonzero rational number r is the power of p in the reduced form of r. For example,

$$\frac{40}{7} = \frac{2^3 \cdot 5}{7} = 2^3 \cdot 5^1 \cdot 7^{-1}$$

so we say $40/7$ has 2-multiplicity 3, 5-multiplicity 1, 7-multiplicity -1, and p-multiplicity 0 for primes p other than 2, 5, and 7. Denote the p-multiplicity of r as $m_p(r)$, so the above calculations say $m_2(40/7) = 3$, $m_5(40/7) = 1$, $m_7(40/7) = -1$, and $m_p(40/7) = 0$ for primes p other than 2, 5, and 7.

Since the p-multiplicity is an exponent, it behaves like a logarithm:

$$m_p(ab) = m_p(a) + m_p(b), \quad m_p(a/b) = m_p(a) - m_p(b). \tag{5.1}$$

For example, $m_2(80) = m_2(16 \cdot 5) = 4$ and $m_2(8) + m_2(10) = 3 + 1 = 4$, while $m_2(80/14) = m_2(40/7) = 3$ and $m_2(80) - m_2(14) = 4 - 1 = 3$. The rules (5.1) essentially come from unique prime factorization in $\mathbf{Z}$.

A nonzero rational number is an integer exactly when its p-multiplicity is nonnegative for *all* primes p (a rational number that is not an integer has negative p-multiplicity for primes p appearing in its reduced form denominator). We will prove $\binom{n}{k} \in \mathbf{Z}$. by showing $m_p(\binom{n}{k}) \geq 0$ for every prime p. According to [2, p. 263], this argument is due to André [1, pp. 188-189].

For prime p and integer $N \geq 0$, Legendre [3, p. 10] proved a formula for $m_p(N!)$:

$$m_p(N!) = \sum_{i \geq 1} \left\lfloor \frac{N}{p^i} \right\rfloor = \left\lfloor \frac{N}{p} \right\rfloor + \left\lfloor \frac{N}{p^2} \right\rfloor + \left\lfloor \frac{N}{p^3} \right\rfloor + \cdots,$$

where the sum is finite since the ith term is 0 once $p^i > N$.[2] Thus

$$\begin{aligned} m_p\left(\binom{n}{k}\right) &= m_p\left(\frac{n!}{k!(n-k)!}\right) \\ &= m_p(n!) - m_p(k!) - m_p((n-k)!) \\ &= \sum_{i \geq 1} \left\lfloor \frac{n}{p^i} \right\rfloor - \sum_{i \geq 1} \left\lfloor \frac{k}{p^i} \right\rfloor - \sum_{i \geq 1} \left\lfloor \frac{n-k}{p^i} \right\rfloor \\ &= \sum_{i \geq 1} \left(\left\lfloor \frac{n}{p^i} \right\rfloor - \left\lfloor \frac{k}{p^i} \right\rfloor - \left\lfloor \frac{n-k}{p^i} \right\rfloor \right). \end{aligned} \tag{5.2}$$

Each term in this last sum has the form $\lfloor x+y \rfloor - \lfloor x \rfloor - \lfloor y \rfloor$, where $x = k$ and $y = n-k$. For all real numbers x and y, $\lfloor x+y \rfloor$ is either $\lfloor x \rfloor + \lfloor y \rfloor$ or $\lfloor x \rfloor + \lfloor y \rfloor + 1$: if the decimal parts of x and y have sum in $[0,1)$ then $\lfloor x+y \rfloor = \lfloor x \rfloor + \lfloor y \rfloor$, while if the decimal parts of x and y have sum in $[1,2)$ then $\lfloor x+y \rfloor = \lfloor x \rfloor + \lfloor y \rfloor + 1$.

[2] A second formula for $m_p(N!)$, also due to Legendre [3, p. 12], is $(N - s_p(N))/(p-1)$, where $s_p(N)$ is the sum of the base p digits of N.

Since $\lfloor x+y\rfloor - \lfloor x\rfloor - \lfloor y\rfloor$ is always 0 or 1, the formula (5.2) shows $m_p(\binom{n}{k})$ is a sum of terms that are each 0 or 1. Thus $m_p(\binom{n}{k}) \geq 0$ for every prime p, so $\binom{n}{k} \in \mathbf{Z}^+$.

6. Proof by Group Theory

For a subgroup H of a finite group G, $|G|/|H|$ is a positive integer by Lagrange's theorem: $|G|/|H|$ is the index $[G:H]$. We'll use this for $G = S_n$, of order $n!$. For $1 \leq k \leq n-1$, the permutations of $\{1,2,\ldots,n\}$ carrying $\{1,\ldots,k\}$ and $\{k+1,\ldots,n\}$ back to themselves form a subgroup H isomorphic to $S_k \times S_{n-k}$, so $|H| = k!(n-k)!$ and $\binom{n}{k} = [G:H]$.[3]

7. Proof by Polynomial Functions

For a variable x, set

$$\binom{x}{k} = \frac{x(x-1)\cdots(x-(k-1))}{k!} = \frac{x(x-1)\cdots(x-k+1)}{k!}.$$

This is a polynomial of degree k. For example,

$$\binom{x}{0} = 1, \quad \binom{x}{1} = x, \quad \binom{x}{2} = \frac{x^2-x}{2}, \quad \binom{x}{3} = \frac{x^3-3x^2+2x}{6}.$$

We can see that the coefficients are not generally integers.

For an integer $n \geq k$, the value of $\binom{x}{k}$ at $x = n$ is $\binom{n}{k}$. Since we have a polynomial, we can substitute in values of x that are integers less than k. When $k \geq 1$ and $x = 0, 1, \ldots, k-1$ we have $\binom{x}{k} = 0$. At $x = k$, the value of $\binom{x}{k}$ is 1. This information turns out to be enough to deduce that the values of $\binom{x}{k}$ at larger integers are also in $\mathbf{Z}$.

Theorem 7.1. *If $f(x)$ is a polynomial of degree k and $f(0), f(1), \ldots, f(k)$ are in $\mathbf{Z}$ then $f(n) \in \mathbf{Z}$ for every integer $n \geq 0$.*

Proof. We induct on $\deg f(x)$. The result is obvious if $k = 0$. For $k = 1$, $f(x) = ax + b$ for a and b in $\mathbf{Q}$, so $f(0) = b$ and $f(1) = a + b$. Thus $a = f(1) - b = f(1) - f(0)$. and $b = f(0)$ are in $\mathbf{Z}$, so $f(n) = an + b$ is in $\mathbf{Z}$ when n is a nonnegative integer (or an arbitrary integer).

Now suppose $\deg f = k \geq 2$ and the theorem is proved for polynomials of degree less than k. Set $\boxed{g(x) = f(x+1) - f(x)}$. What does this look like? Writing $f(x) = a_k x^k + a_{k-1}x^{k-1} + \ldots + a_1 x + a_0$, we have

$$\begin{aligned} g(x) &= f(x+1) - f(x) \\ &= (a_k(x+1)^k + a_{k-1}(x+1)^{k-1} + \cdots + a_0) - (a_k x^k + a_{k-1}x^{k-1} + \cdots + a_0) \\ &= a_k((x+1)^k - x^k) + a_{k-1}((x+1)^{k-1} - x^{k-1}) + \cdots + a_1(x+1-x) \\ &= ka_k x^{k-1} + \text{ lower-degree terms}, \end{aligned}$$

where the last formula is due to $(x+1)^k = x^k + kx^{k-1} +$ lower-degree terms. (This is a weak form of the binomial theorem.) Since $a_k \neq 0$, $g(x)$ has degree $k-1$.

From $f(0), f(1), \ldots, f(k)$ being in $\mathbf{Z}$, the values $g(0) = f(1) - f(0)$, $g(1) = f(2) - f(1)$, $\ldots$, $g(k-1) = f(k) - f(k-1)$ are in $\mathbf{Z}$ since each one is a difference of integers. By induction

[3] Also $\frac{n!}{k!(n-k)!} \in \mathbf{Z}^+$ since the k-element subsets of $\{1,\ldots,n\}$ are an orbit of S_n with stabilizer $S_k \times S_{n-k}$.

on polynomial degrees, $g(n) \in \mathbf{Z}$ for all nonnegative integers n. Then for integers $n \geq 1$,

$$\begin{aligned} f(n) &= (f(n) - f(n-1)) + (f(n-1) - f(n-2)) + \cdots + (f(1) - f(0)) + f(0) \\ &= g(n-1) + g(n-2) + \cdots + g(0) + f(0), \end{aligned} \tag{7.1}$$

which is a sum of integers, so $f(n) \in \mathbf{Z}$. □

Remark 7.2. The theorem admits a slightly stronger conclusion than we have shown: $f(n) \in \mathbf{Z}$ for all $n \in \mathbf{Z}$, not just when $n \geq 0$.

This approach to the integrality of binomial coefficients shares a lot in common with the proof by recursion in Section 3. Indeed, for $k \geq 1$ the polynomial $\binom{x}{k}$ satisfies $\binom{x+1}{k} = \binom{x}{k-1} + \binom{x}{k}$, so if $f(x) = \binom{x}{k}$ then $g(x) := f(x+1) - f(x)$ is $\binom{x}{k-1}$. In this case, (7.1) says

$$\begin{aligned} \binom{n}{k} &= g(n-1) + g(n-2) + \cdots + g(0) + f(0) \\ &= \binom{n-1}{k-1} + \binom{n-2}{k-1} + \cdots + \binom{0}{k-1} + \binom{0}{k}. \end{aligned}$$

The term $f(0)$ is 0, and also $g(j) = 0$ for $j = 0, \ldots, k-2$, so (7.1) for $f(x) = \binom{x}{k}$ says

$$\binom{n}{k} = g(n-1) + g(n-2) + \cdots + g(k-1) = \sum_{j=k-1}^{n-1} \binom{j}{k-1},$$

which is exactly (3.2).

8. q-BINOMIAL COEFFICIENTS

For a real number $q > 0$ with $q \neq 1$, or an indeterminate q, the positive integer n can be generalized to the polynomial

$$(n)_q = \frac{q^n - 1}{q - 1} = 1 + q + \cdots + q^{n-1}.$$

This polynomial is called the q-*analogue* of n, and at $q = 1$ its value is n. For example,

$$(1)_q = 1, \quad (2)_q = 1 + q, \quad (3)_q = 1 + q + q^2.$$

The expression of $(n)_q$ as the ratio $(q^n - 1)/(q - 1)$ does not make direct sense at $q = 1$, but its limit as $q \to 1$ is n. The product

$$(n)_q! = (n)_q (n-1)_q \cdots (2)_q (1)_q,$$

is called the q-*factorial* of n, and we set $(0)_q! = 1$ (analogous to setting $0! = 1$). The ratio

$$\binom{n}{k}_q := \frac{(n)_q!}{(k)_q!(n-k)_q!} = \frac{(n)_q (n-1)_q \cdots (n-k+1)_q}{(k)_q!} \tag{8.1}$$

is called a q-*binomial coefficient*. Each $(j)_q$ is a polynomial in q, so $\binom{n}{k}_q$ is a rational function of q. Since $(n)_q! = n!$ at $q = 1$, we have $\binom{n}{k}_q = \binom{n}{k}$ at $q = 1$.

The second defining formula for $\binom{n}{k}_q$ in (8.1) has an equal number of factors in the numerator and denominator (the first defining formula in (8.1) does as well), so writing $(j)_q = (q^j - 1)/(q - 1)$ gives us a third formula for q-binomial coefficients:

$$\binom{n}{k}_q := \frac{(q^n - 1)(q^{n-1} - 1) \cdots (q^{n-k+1} - 1)}{(q^k - 1)(q^{k-1} - 1) \cdots (q - 1)}. \tag{8.2}$$

Example 8.1. Since $(0)_q! = 1$ and $(1)_q! = 1$, we have $\binom{n}{0}_q = \binom{n}{n}_q = 1$ for $n \geq 0$ and $\binom{n}{1}_q = \binom{n}{n-1}_q = (n)_q = 1 + q + \cdots + q^{n-1}$ for $n \geq 1$, which generalize $\binom{n}{0} = \binom{n}{n} = 1$ for $n \geq 0$ and $\binom{n}{1} = \binom{n}{n-1} = n$ for $n \geq 1$.

Example 8.2. The first q-binomial coefficient $\binom{n}{k}_q$ with $k \neq 0, 1, n-1$, or n is

$$\binom{4}{2}_q = \frac{(4)_q(3)_q}{(2)_q!} = \frac{(1+q+q^2+q^3)(1+q+q^2)}{(1+q)} = (1+q^2)(1+q+q^2),$$

where the last formula comes from the factorization $1+q+q^2+q^3 = (1+q)(1+q^2)$. Setting $q = 1$, we recover the value $\binom{4}{2} = 2 \cdot 3 = 6$.

The calculation in Example 8.2 is a special case of the following general theorem.

Theorem 8.3. *For integers $n \geq k \geq 0$, $\binom{n}{k}_q$ is a polynomial in q with coefficients that are nonnegative integers.*

Theorem 8.3 implies the integrality of binomial coefficients by setting $q = 1$ in the conclusion of Theorem 8.3. We will show how the ideas in most of the proofs of integrality of $\binom{n}{k}$ can be adapted to give a proof of Theorem 8.3, sometimes in the weaker form that the coefficients are in $\mathbf{Z}$ rather than being nonnegative in $\mathbf{Z}$. That weaker form suffices to see that $\binom{n}{k} = \binom{n}{k}_1$ is an integer, and it is clearly positive from its definition.

Proof of Theorem 8.3 by combinatorics:

The q-analogue of $(1+x)^n$ is $(1+x)(1+qx)(1+q^2x)\cdots(1+q^{n-1}x)$. By distributivity,

$$(1+x)(1+qx)\cdots(1+q^{n-1}x) = \sum_{k=0}^{n} c_{n,k}(q)x^k \tag{8.3}$$

where $c_{n,k}(q)$ is a polynomial in q with nonnegative integer coefficients. Clearly $c_{n,0}(q) = 1$ and $c_{n,n}(q) = q^{0+1+2+\cdots+(n-1)} = q^{n(n-1)/2}$. For $q \neq 1$ we will get a formula for $c_{n,k}(q)$ by computing the left side of (8.3) in two ways. First,

$$\begin{aligned}
(1+x)(1+qx)\cdots(1+q^nx) &= (1+x)(1+qx)\cdots(1+q^{n-1}x)\cdot(1+q^nx) \\
&= \left(\sum_{k=0}^{n} c_{n,k}(q)x^k\right)(1+q^nx) \\
&= \sum_{k=0}^{n} c_{n,k}(q)x^k + \sum_{k=0}^{n} q^n c_{n,k}(q)x^{k+1} \\
&= 1 + \sum_{k=1}^{n} c_{n,k}(q)x^k + \sum_{k=1}^{n} q^n c_{n,k-1}(q)x^k + q^{n+n(n-1)/2}x^{n+1} \\
&= 1 + \sum_{k=1}^{n} (c_{n,k}(q) + q^n c_{n,k-1}(q))x^k + q^{n+n(n-1)/2}x^{n+1}.
\end{aligned}$$

Second,

$$\begin{aligned}(1+x)(1+qx)\cdots(1+q^nx) &= (1+x)\cdot(1+qx)\cdots(1+q^{n-1}x)(1+q^nx)\\ &= (1+x)\left(\sum_{k=0}^{n} c_{n,k}(q)(qx)^k\right)\\ &= 1+\sum_{k=1}^{n}\left(q^kc_{n,k}(q)+q^{k-1}c_{n,k-1}(q)\right)x^k + q^{n(n-1)/2+n}x^{n+1}\end{aligned}$$

Equating coefficients of x^k in both formulas, for $1 \leq k \leq n$,

$$c_{n,k}(q)+q^nc_{n,k-1}(q) = q^kc_{n,k}(q)+q^{k-1}c_{n,k-1}(q) \Longrightarrow c_{n,k}(q) = \frac{q^n-q^{k-1}}{q^k-1}c_{n,k-1}(q).$$

(If we take the limit as $q \to 1$, this recursive formula becomes (2.1).) Iterating this recursion $k-1$ more times,

$$\begin{aligned}c_{n,k}(q) &= \frac{(q^n-q^{k-1})(q^n-q^{k-2})\cdots(q^n-1)}{(q^k-1)(q^{k-1}-1)\cdots(q-1)}c_{n,0}(q)\\ &= \frac{q^{k-1}(q^{n-(k-1)}-1)q^{k-2}(q^{n-(k-2)}-1)\cdots q^0(q^n-1)}{(q^k-1)(q^{k-1}-1)\cdots(q-1)}\\ &= \frac{q^{(k-1)+(k-2)+\cdots+1}(q^n-1)(q^{n-1}-1)\cdots(q^{n-k+1}-1)}{(q^k-1)(q^{k-1}-1)\cdots(q-1)}\\ &= q^{k(k-1)/2}\binom{n}{k}_q,\end{aligned}$$

where the last formula comes from (8.2). This proves $q^{k(k-1)/2}\binom{n}{k}_q$ is a polynomial in q with nonnegative integer coefficients when $q \neq 1$. To conclude that $\binom{n}{k}_q$ is also a polynomial in q, here are two approaches. Each one treats the previous calculations as if q is an indeterminate.

(1) In the expansion (8.3), $c_{n,k}(q)$ is a sum of products of distinct q-powers k at a time, and each of those products has exponent at least $0+1+\cdots+(k-1) = k(k-1)/2$. Therefore $c_{n,k}(q)$ is a polynomial in q that's divisible by $q^{k(k-1)/2}$, so $\binom{n}{k}_q$ is a polynomial in q.

(2) By (8.2), $\binom{n}{k}_q$ is a rational function of q with no q-power in its denominator, so when its product with a power of q is a polynomial in q, $\binom{n}{k}_q$ is also a polynomial in q.

Proof of Theorem 8.3 by recursion:

A recursion for q-binomial coefficients that generalizes the Pascal's triangle recursion for binomial coefficients is

$$\binom{n}{k}_q = q^{n-k}\binom{n-1}{k-1}_q + \binom{n-1}{k}_q.$$

Iterating this enough times, we get

$$\binom{n}{k}_q = \sum_{j=0}^{n-k} q^{n-k-j}\binom{n-1-j}{k-1}_q = \sum_{j=k-1}^{n-1} q^{j-(k-1)}\binom{j}{k-1}_q,$$

which generalizes (3.2). If all q-binomial coefficients with denominator $k-1$ are polynomials in q with nonnegative integer coefficients then so is each $\binom{n}{k}_q$ by the above formula. Therefore starting from the value $\binom{0}{0}_q = 1$, Theorem 8.3 is proved by induction on k.

Proof of Theorem 8.3 by calculus:

A q-analogue of the series identity

$$\frac{1}{(1-x)^m} = \sum_{k\geq 0} \binom{m+k-1}{k} x^k$$

for $|x| < 1$ and $m \geq 1$ is

$$\frac{1}{(1-x)(1-qx)\cdots(1-q^{m-1}x)} = \sum_{k\geq 0} \binom{m+k-1}{k}_q x^k$$

for $|x| < 1$, $0 < q < 1$, and $m \geq 1$. Since each $1/(1-q^i x)$ for $|x| < 1$ and $0 < q < 1$ can be expanded into a geometric series $\sum_{k\geq 0} q^{ik}x^k$, the product of these series is a power series in x having coefficients that are polynomials in q with nonnegative integer coefficients. Therefore $\binom{m+k-1}{k}_q$ is a polynomial in q with nonnegative integer coefficients, at least when $0 < q < 1$. A polynomial and rational function in q that agree for $0 < q < 1$ must agree for all q, so we get Theorem 8.3 using $m = n-k+1$.

Proof of Theorem 8.3 by number theory:

Treat q as an indeterminate. By (8.2), $\binom{n}{k}_q$ is a ratio of products of polynomials of the form $q^i - 1$, so the irreducible factors in the numerator and denominator of (8.2) are cyclotomic polynomials $\Phi_j(q)$ for various j. We will show every cyclotomic polynomial $\Phi_j(q)$ has nonnegative multiplicity in $\binom{n}{k}_q$, so $\binom{n}{k}_q$ is a product of cyclotomic polynomials and therefore is a polynomial with integral coefficients. (This will not tell us the coefficients of that polynomial in q are nonnegative.) For positive integers i and j, $\Phi_j(q)$ is a factor of $q^i - 1$ if and only if $j \mid i$, so the multiplicity of $\Phi_j(q)$ as a factor of the polynomial $(n)_q!$ is $|\{i \leq n : j \mid i\}| = \lfloor n/j \rfloor$. Therefore the multiplicity of $\Phi_j(q)$ in $\binom{n}{k}_q = (n)_q!/(k)_q!(n-k)_q!$ is $\lfloor n/j \rfloor - \lfloor k/j \rfloor - \lfloor (n-k)/j \rfloor$, which has the form $\lfloor x+y \rfloor - \lfloor x \rfloor - \lfloor y \rfloor$, so it is 0 or 1. (Thus no $\binom{n}{k}_q$ has a repeated irreducible factor, which is a contrast to ordinary binomial coefficients $\binom{n}{k}$ since they often have repeated prime factors. In fact, the largest n for which all $\binom{n}{k}$ are squarefree is 23. See http://oeis.org/A048278.)

Proof of Theorem 8.3 by group theory:

Let $q = p$ be an arbitrary prime number. We will use group theory to show $\binom{n}{k}_p$ is a positive integer and then explain why this implies $\binom{n}{k}_q$ is a polynomial in q for general q.

The group $\mathrm{GL}_n(\mathbf{F}_p)$ is all the automorphisms of the additive group $\mathbf{F}_p^n$. Using linear algebra (counting bases of $\mathbf{F}_p^n$), the order of this group is

$$\begin{aligned}(p^n-1)(p^n-p)\cdots(p^n-p^{n-1}) &= (p^n-1)(p^{n-1}-1)p\cdots(p-1)p^{n-1}\\ &= (p^n-1)(p^{n-1}-1)\cdots(p-1)p^{n(n-1)/2}\\ &= (n)_p!p^{n(n-1)/2}.\end{aligned}$$

Inside $\mathrm{GL}_n(\mathbf{F}_p)$, the matrices A that preserve the direct sum decomposition $\mathbf{F}_p^k \oplus \mathbf{F}_p^{n-k}$ (this means A carries $\mathbf{F}_p^k \oplus \mathbf{0}$ to itself and $\mathbf{0} \oplus \mathbf{F}_p^{n-k}$ to itself) is a subgroup. This subgroup is isomorphic to $\mathrm{GL}_k(\mathbf{F}_p) \times \mathrm{GL}_{n-k}(\mathbf{F}_p)$, so its order is $(k)_p!p^{k(k-1)/2}(n-k)_p!p^{(n-k)(n-k-1)/2}$.

By Lagrange's theorem, the ratio

$$\frac{|\,\mathrm{GL}_n(\mathbf{F}_p)|}{|\,\mathrm{GL}_k(\mathbf{F}_p)\times \mathrm{GL}_{n-k}(\mathbf{F}_p)|} = \frac{(n)_p!p^{n(n-1)/2}}{(k)_p!p^{k(k-1)/2}(n-k)_p!p^{(n-k)(n-k-1)/2}}$$
$$= \frac{(n)_p!}{(k)_p!(n-k)_p!}p^{k(n-k)}$$
$$= \binom{n}{k}_p p^{k(n-k)}$$

is an integer. Since $(n)_p!$ and $(k)_p!(n-k)_p!$ are integers that are not divisible by p, we can conclude from $\binom{n}{k}_p p^{k(n-k)} \in \mathbf{Z}$ that $\binom{n}{k}_p \in \mathbf{Z}$.

If a rational function in an indeterminate q has numerator and denominator that are polynomials in q with coefficients in $\mathbf{Z}$, and its values are in $\mathbf{Z}$ as q runs through the prime numbers, then the rational function is a polynomial in q with coefficients in $\mathbf{Q}$. This is a special case of Theorem A.1 below, and it tells us $\binom{n}{k}_q$ is a polynomial in q. But watch out: a polynomial with coefficients in $\mathbf{Q}$ that has integral values at primes need not have coefficients in $\mathbf{Z}$. For example, $\frac{1}{2}(q^2-q)$ has values in $\mathbf{Z}$ when q runs through $\mathbf{Z}$ (not just primes) but its coefficients are not in $\mathbf{Z}$. To show $\binom{n}{k}_q$ has coefficients in $\mathbf{Z}$, we can use more information about q-binomial coefficients: they are a ratio of monic polynomials with integer coefficients, by (8.2). It can be shown (see Theorem A.1) that a polynomial with coefficients in $\mathbf{Q}$ that is equal to a ratio of monic polynomials with coefficients in $\mathbf{Z}$ must itself have coefficients in $\mathbf{Z}$. This is why $\binom{n}{k}_q$ has coefficients in $\mathbf{Z}$, but we don't learn why the coefficients are nonnegative.

Remark 8.4. The proof of integrality of $\binom{n}{k}$ in Section 7, using polynomial functions and the discrete difference operation $f(x+1)-f(x)$ to reduce degrees, has a q-analogue: there are q-discrete difference operations that reduce "q-degrees" (*e.g.*, the sequence $f(n)=q^{kn}$ for each $k \geq 0$ has q-degree k). We have already given enough other proofs of Theorem 8.3 that we omit the proof analogous to the method in Section 7.

Appendix A. A criterion for a rational function to be a polynomial

We used the theorem below in the proof of Theorem 8.3 by group theory.

Theorem A.1. *If $A(x)$ and $B(x)$ are polynomials with coefficients in $\mathbf{Z}$ and $A(n)/B(n) \in \mathbf{Z}$ for infinitely many integers n, then $A(x)/B(x)$ is a polynomial with rational coefficients.*

If $B(x)$ has leading coefficient 1 then $A(x)/B(x)$ is a polynomial with integral coefficients.

This is *not* saying $B(x)$ is constant, since $A(x)/B(x)$ may not be reduced at first. For example, $(x^4-1)/(x^2-1)$ and $(x^4-1)/(x^2+1)$ fits the conditions of the theorem.

Proof. The result is obvious if $A(x)$ is 0, so assume $A(x) \neq 0$. Polynomials with integral coefficients have unique factorization, so we can suppose $A(x)/B(x)$ is in reduced form. Then $A(x)$ and $B(x)$ have no nonconstant common factor in $\mathbf{Q}[x]$, so $\gcd(A(x),B(x))=1$ in $\mathbf{Q}[x]$. By Bezout's identity, $A(x)U(x)+B(x)V(x)=1$ for some $U(x),V(x) \in \mathbf{Q}[x]$. Multiplying through by a common denominator of the coefficients of $U(x)$ and $V(x)$, there are $\widetilde{U}(x)$ and $\widetilde{V}(x)$ in $\mathbf{Z}[x]$ and $c \in \mathbf{Z}-\{0\}$ such that

$$A(x)\widetilde{U}(x)+B(x)\widetilde{V}(x)=c. \tag{A.1}$$

Let S be the set of $n \in \mathbf{Z}$ such that $A(n)/B(n) \in \mathbf{Z}$, so S is infinite. Thus S contains n with $|n| \to \infty$. For $n \in S$ we have $B(n) \mid A(n)$, so (A.1) implies $B(n) \mid c$. If $B(x)$ is nonconstant, then $|B(n)| \to \infty$ as $|n| \to \infty$, but a nonzero integer doesn't have factors that have arbitrarily large absolute value. Therefore $B(x)$ is constant, so $A(x)/B(x) \in \mathbf{Q}[x]$.

Suppose now that $B(x)$ has leading coefficient 1. Then we can divide $A(x)$ by $B(x)$ using polynomials with integral coefficients: $A(x) = B(x)q(x) + r(x)$ where $q(x)$ and $r(x)$ have integral coefficients and $r(x) = 0$ or $\deg r < \deg B$. Since $B(x) \mid A(x)$ in $\mathbf{Q}[x]$, the uniqueness of quotient and remainder in $\mathbf{Q}[x]$ implies $r(x) = 0$, so $A(x)/B(x) = q(x)$ is a polynomial with integral coefficients. □

References

[1] D. André, *Sur une formule d'arithmétique*, Nouv. Ann. Math. **13** (1874), 185–189. URL http://www.numdam.org/item/NAM_1874_2_13__185_1/.

[2] L. E. Dickson, "History of the Theory of Numbers, Vol. I: Primality and Divisibility," Chelsea, Bronx, NY, 1971.

[3] A.-M. Legendre, "Théorie des Nombres, Tome 1," 3rd ed., Firmin Didot Frères, Paris, 1830. URL https://books.google.com/books?id=BTIVAAAAQAAJ.